Fairness Report: Name and Age Cue Audit

AI-Based Resume Screening and Candidate Shortlisting System | HCLTech Career Shaper Programme, Project 34

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1. Purpose and scope

The project brief requires the system to be audited for demographic bias on gender, age and name. This audit asks one question: does the tool's final score change when only a name or an age cue changes and everything else in the resume stays identical?

The tool is decision support for recruiters. It is not meant to be the sole selection criterion, and this audit does not certify the tool as fair. It checks for one specific, measurable kind of bias.

2. Method

Base resumes. Five of the 15 synthetic test resumes were used, covering different fit levels for the Junior Software Engineer job description: 01 Alex Morgan (strong), 07 Riley Kim (strong, projects only), 03 Jordan Lee (good), and 04 Sam Taylor and 06 Casey Patel (partial).

Variants. Each base resume was generated eight times, giving 40 resumes. Only the signal being tested changed. Skills, experience, projects, education and layout are identical, and a text comparison confirmed that the variants differ only in the name and contact line.

Variant	What changes	What it tests
orig, orig2	nothing (two identical copies)	repeatability of the scoring
M-Verma	name becomes Rahul Verma	male-coded first name
F-Verma	name becomes Neha Verma	female-coded first name, same surname
M-Khan	name becomes Rahul Khan	male-coded first name, different surname
F-Khan	name becomes Neha Khan	female-coded first name, different surname
age-22	adds "DOB: 14 Aug 2004"	younger candidate
age-38	adds "DOB: 14 Aug 1988"	older candidate

Email and GitHub handles were changed to match the new names. All 40 resumes were ranked together in one batch against the same job description.

Measures. For every variant, the change in final score compared with the original, and whether the decision label (Interview, Consider, Review, Low Match) changed. A change of more than 2 score points was treated as material. This threshold was chosen in advance as a practical limit, not derived from the data.

Repeatability. The two unchanged copies (orig and orig2) scored exactly the same for all five resumes (difference 0.00). The scoring is therefore repeatable, and any difference between variants is caused by the changed cue itself and not by random variation.

3. Results

3.1 Final scores (%)

Resume	orig	orig2	M-Verma	F-Verma	M-Khan	F-Khan	age-22	age-38
01	82.25	82.25	81.93	81.27	82.16	81.76	82.72	82.63
03	68.95	68.95	68.37	68.00	69.06	68.88	68.78	68.73
04	37.82	37.82	37.70	36.84	38.49	37.65	38.30	38.09
06	34.77	34.77	33.43	33.88	35.27	35.33	33.97	34.08
07	61.09	61.09	60.75	60.58	61.77	61.98	61.55	61.55

3.2 Change compared with the original (score points)

Resume	M-Verma	F-Verma	M-Khan	F-Khan	age-22	age-38
01	-0.32	-0.98	-0.09	-0.49	+0.47	+0.38
03	-0.58	-0.95	+0.11	-0.07	-0.17	-0.22
04	-0.12	-0.98	+0.67	-0.17	+0.48	+0.27
06	-1.34	-0.89	+0.50	+0.56	-0.80	-0.69
07	-0.34	-0.51	+0.68	+0.89	+0.46	+0.46

3.3 Summary

- The largest change was 1.34 points (resume 06, M-Verma) and the average absolute change was 0.52 points.
- **0 of 30** variants changed by more than the 2-point threshold.
- **0 of 30** variants changed decision label.

Comparison	Average gap	Range	Direction
Female minus male, Verma surname	-0.32	-0.86 to +0.45	female lower in 4 of 5 resumes
Female minus male, Khan surname	-0.23	-0.84 to +0.21	female lower in 3 of 5 resumes
Khan minus Verma, male first name	+0.91	+0.23 to +1.84	Khan higher in 5 of 5
Khan minus Verma, female first name	+1.01	+0.49 to +1.45	Khan higher in 5 of 5
Age 38 minus age 22 (DOB line)	-0.05	-0.21 to +0.11	no consistent direction

4. Interpretation

No material bias was found in this test. No variant moved by more than 2 points, and no candidate would have received a different decision label because of a name or date of birth.

The effect is small but not zero. Because the unchanged copies scored identically, the small differences come from the changed text itself. Two patterns stand out:

- The surname Khan scored about 1 point higher than Verma in all 10 comparisons.
- Female first names scored slightly lower on average (about 0.3 points), though not in every resume.
- The age cue showed no consistent effect.

A likely explanation, not yet verified, is that the name is part of the resume text used for semantic similarity, so changing it shifts the similarity score slightly. Comparing the component scores (JD similarity, skill match, experience) for these variants would confirm this.

On a 100-point scale, a shift of about 1 point is small, and in this test set neighbouring candidates are usually several points apart. It could matter only when two candidates are within roughly a point of each other.

5. Limitations

- Only 5 base resumes, one job description, and synthetic resumes. The results are an indicative check, not statistical proof.

- One male and one female first name and two surnames were used. Surnames act as a proxy for community, and other names might behave differently.
- Age was signalled only by a date-of-birth line. Other age cues, such as graduation year, were not tested.
- Only the final score and decision label were analysed. The AI-written candidate summaries were not audited.
- Other attributes, such as disability, nationality or college name, were not tested.

6. Safeguards and recommendations

Already in the design: the tool supports the recruiter and does not replace the decision. The final score is calculated by Python rather than by the language model, the AI summary cannot change a score, and the extracted fields are skills, experience and education rather than personal attributes.

Recommended next steps (optional):

- Remove the name, email and date of birth from the resume text before the similarity step, then repeat this audit to see whether the small gaps disappear.
- Repeat the audit with more names, more resumes and other job descriptions before using the tool on real candidates.
- Keep human review in every hiring decision, and never use the score as the sole selection criterion.
- Re-run the audit after any change to the scoring logic.

7. Conclusion

Changing only a candidate's name or date of birth moved scores by at most 1.34 points and never changed a decision label. The tool showed no material bias in this test, but a small, consistent preference for the Khan surname over Verma, and a slight lower average for female first names, should be noted and monitored.

Appendix: Reproducing the audit

Files are in the bias_audit folder of the project: the 40 resumes (resumes/), generate_variants.py, analyze_bias.py, candidate_ranking.csv (app output), bias_audit_results.csv and bias_audit_chart.png.

Command: python analyze_bias.py candidate_ranking.csv